



PDF Download
3716640.3716655.pdf
16 January 2026
Total Citations: 1
Total Downloads: 1579

Latest updates: <https://dl.acm.org/doi/10.1145/3716640.3716655>

RESEARCH-ARTICLE

Leveling up Learning: Serious Games for Computing Education - Long-Term Opportunities and Risks

BURKHARD CLAUS WÜNSCHE, University of Auckland, Auckland, AUK, New Zealand

STEFFAN HOOPER, University of Auckland, Auckland, AUK, New Zealand

JACQUELINE L. WHALLEY, Auckland University of Technology, Auckland, AUK, New Zealand

INGJERD J STRAAND, University of Stavanger, Stavanger, Norway

PAUL DENNY, University of Auckland, Auckland, AUK, New Zealand

TYNE CROW, University of Auckland, Auckland, AUK, New Zealand

[View all](#)

Open Access Support provided by:

[Auckland University of Technology](#)

[University of Auckland](#)

[University of Stavanger](#)

[University of Cambridge](#)

Published: 12 February 2025

[Citation in BibTeX format](#)

ACE 2025: 27th Australasian Computing
Education Conference
February 12 - 13, 2025
Brisbane, Australia

Leveling up Learning: Serious Games for Computing Education - Long-Term Opportunities and Risks

Burkhard Claus Wünsche
The University of Auckland
Auckland, New Zealand
burkhard@cs.auckland.ac.nz

Steffan Hooper
University of Auckland
Auckland, New Zealand
steffan.hooper@auckland.ac.nz

Jacqueline Whalley
Auckland University of Technology
Auckland, New Zealand
jacqueline.whalley@aut.ac.nz

Ingjerd Straand
University of Stavanger
Stavanger, Norway
ingjerd.j.straand@uis.no

Paul Denny
The University of Auckland
Auckland, New Zealand
paul@cs.auckland.ac.nz

Tyne Crow
The University of Auckland
Auckland, New Zealand
t.crow@auckland.ac.nz

Dominik Lange-Nawka
University of Auckland
Auckland, New Zealand
dlan133@aucklanduni.ac.nz

Andrew Luxton-Reilly
University of Auckland
Auckland, New Zealand
andrew@cs.auckland.ac.nz

Samuel E. R. Thompson
University of Cambridge
Cambridge, United Kingdom
sam.thompson.xr@gmail.com

Abstract

Serious games are becoming an increasingly popular tool in education, including computing education, due to their potential to engage learners, practice skills development, stimulate creativity, promote active learning, and cultivate exploration. However, few researchers have discussed the potential long-term implications of using serious games for learning programming and how they can be addressed.

This paper investigates opportunities and drawbacks arising from the wider use of serious games in computing education. We look beyond immediate effects, such as the risk of distraction and technological barriers, and consider prospects and risks caused by market forces and the economic realities of the game, education, and software industries. In particular, we discuss how technological advances can increase the benefits of serious games for programming education, but also that there is a risk that an increasing use of serious games might result in market dominance of a few vendors, lack of diversity, cultural bias, inequity, pressure on education funding, and reduced learning outcomes.

CCS Concepts

• **Software and its engineering** → **Interactive games**; • **Applied computing** → **Computer games**; **Education**.

Keywords

Serious Games, Computing Education, Diversity, Equality, Accessibility, Generative AI, AR/VR, Game Industry

ACM Reference Format:

Burkhard Claus Wünsche, Steffan Hooper, Jacqueline Whalley, Ingjerd Straand, Paul Denny, Tyne Crow, Dominik Lange-Nawka, Andrew Luxton-Reilly, and Samuel E. R. Thompson. 2025. Leveling up Learning: Serious Games for Computing Education - Long-Term Opportunities and Risks. In *27th Australasian Computing Education Conference (ACE 2025)*, February 12–13, 2025, Brisbane, QLD, Australia. ACM, New York, NY, USA, 10 pages. <https://doi.org/10.1145/3716640.3716655>

1 Introduction

One of the key challenges in computing education is to motivate learners and make the learning process engaging and enjoyable [28, 103]. An increasingly popular approach to achieve this, especially for younger learners, is gamification and serious games [74]. This is distinct from the use of games as a context to teach computing, such as the development of video games as programming exercises in introductory courses [72]. Serious games are video games designed for a primary purpose other than pure entertainment [36].

In 2021 the global gamification in education market was valued at US\$860 million and is projected to surpass US\$11 billion in 2030 [114]. 74% of teachers use digital games for instructional purposes with their students, and 87% of teachers found that students were more engaged when including purposeful play [114]. Hence, it is important to understand the opportunities and risks associated with this technological innovation.

In this paper, we review the current state-of-the-art of serious games for computing education and investigate the long-term opportunities and risks of serious games by using a holistic approach encompassing technological innovation, economic considerations, and changes in the education landscape. We will provide arguments that there is promise that serious games can increase the participation and achievements of learners, especially by incorporating new technologies, but that there are also significant risks of unintended consequences, which are currently rarely discussed in the literature. In particular, we argue that there is concern that an increasing use



This work is licensed under a Creative Commons Attribution International 4.0 License.

ACE 2025, Brisbane, QLD, Australia

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1425-2/25/02

<https://doi.org/10.1145/3716640.3716655>

of serious games might result in market dominance of a few vendors, lack of diversity, cultural bias, inequity, pressure on education funding, and reduced learning outcomes.

2 Background

2.1 History of Games in Education

Games are an important part of all cultures and one of the oldest forms of human social interaction. While the primary purpose of games is entertainment, they have been used for centuries for other purposes. Games can be used for cultural and social bonding, e.g., as “ice breaker games” when meeting new people. Many cultures have used games to gain political power, e.g., in the Roman empire games (“bread and circuses”) were used to satisfy the most immediate needs of a populace and to distract from societal, administrative, and political problems [19].

One of the earliest examples of games used as a teaching tool is the game “Go” (“Weiqi”), which appears in records as early as 500 BC. It was used to teach strategic thinking and is claimed to even now still influence Chinese foreign policy [89]. The game “Snakes and ladders” originates from the ancient Indian game “Moksha Patam”, which was used by religious leaders to teach the differences between good and evil – climbing up the ladders represents virtues and snakes represent vices [75]. A more recent example is the game “Monopoly”, the world’s most popular board game with more than 275 million sets sold. It originates from the “Landlord game” created by the American game designer Elizabeth Magie to teach the dangers of land grabbing and the monopolies it creates [70]. The modern version of Monopoly teaches the basic economic concepts of investments, rent, tax, and risks.

We observe that only a small percentage of games have reached a high enough penetration to influence society [23]. Apart from having to be engaging and accessible, the popularity of specific games seems to be driven by the political landscape, economics, and society (needs, beliefs, fashions). For example, it has been claimed that Monopoly became popular because people were looking for an escape from the financial hardships they were feeling and it offered players an opportunity to achieve things they were unlikely to ever achieve in reality [70].

We believe that historic developments indicate that with increasing adoption, the popularity of educational games might be driven by factors other than just their educational value, such as marketing budgets, accessibility (cost), the attractiveness of gameplay, social media, and preferences of decision makers (political interference).

2.2 Psychology and Games

Games incorporate (to different degrees) universal concepts that are important for engagement, i.e., uncertainty, autonomy (choices), skills, achievements, structure (goals/objectives), and social interactions. Learning can be achieved implicitly (“learning by doing”) or explicitly by having a mentor teaching the nuances and/or symbolism of a game [88].

Games are attractive to users because they address the top three levels of Maslow’s hierarchy of needs [53, 77]: (1) Love and Belonging: games encourage social interactions and players often consider themselves a social group; (2) Esteem: most games are easy to play at a basic level, building up player confidence, enable users to improve

skills, and, for games with uncertainty, even weaker players can win occasionally; (3) Self-actualization: most games allow players choices, to explore strategies and game elements, and to grow skills and experience. In addition to these more external focused needs, games may also spur intrinsic motivation when gameplay meets the psychological needs of autonomy, competence and relatedness, as suggested by Self-Determination Theory (SDT) [86, 102, 122].

An important aspect in achieving continuing engagement with a game is replayability. For video games, this can be achieved using a range of strategies offering players a different player experience each time they play the game, e.g., unlocking new characters and items, new levels (manually or procedurally generated), alternative paths and endings (branching narrative), performance grading, or adding restrictions [133]. Replayability requirements for video games are in general different from traditional board games, such as Chess or Monopoly, where the game doesn’t change and replayability is achieved by the complexity of the game, shared physical and social settings, different skills and behaviours of players, and/or chance. Hence replayability of video games is often associated with a large and continuing investment in order to offer players new experiences over an extended period of time (often years) [57].

While replayability is important for long-term engagement, there is conflicting evidence on the effect of replay on students’ learning. Gunter et al. report that replaying a game can help in consolidating learned materials or remediating shortcomings through more practice [49] but other studies found no learning gains [24]. Zhang et al. suggests that the benefits of replay differ for individuals and that replaying previously passed games can be beneficial when failing at a higher, more difficult level in the current game [138]. Vanacore et al. report that in order to improve learning it is important to link replay with gamification elements such as virtual rewards and achievements [124]. This is well supported by research on gamification in education, which has established positive effects of using gamification elements to reward behaviours known to benefit learning [32, 58].

2.3 Educational Games and Economics

The global video games market is projected to reach revenues of US\$282 billion in 2024 with roughly 17% of the global population playing video games [117]. In-game advertising is the largest source of revenue with roughly US\$110bn in 2024, i.e., game companies earn more from advertisements than by selling games. The second largest source of revenue are in-app purchases, which reached US\$75bn in 2022 [117]. This feature is particularly common for Freemium games, which can be downloaded for free, but where players can purchase in-app items to enhance player experience.

Since with recent market trends game revenue depends on advertisements and in-game purchases, there is an incentive for game companies to maximize the player base and play time.

The global gamification in education market was valued at US\$860 million in 2023 and is projected to surpass US\$11 billion in 2030 [114]. Few educational games reach the user base required to generate enough revenue to justify a continuing investment in the platform. We discuss different examples of successful monetization strategies for educational games:

(Perpetual) Licensing: For many decades the traditional market model for games was to sell (perpetual) licenses, which allowed users to use the game indefinitely. The game industry began shifting from perpetual licenses to subscription and freemium models around the mid-2000s due to the rise of digital distribution (Steam, App Store), mobile gaming, desire for constant player engagement and a continuous and growing revenue stream, ability to update content, and market competition [35, 123].

We are not aware of any major serious games for computing education being sold by perpetual license. However, many smaller games (niche products) are available via one-off payments from various digital distribution platforms.

Subscriptions: Arguably the most successful educational video game is “Minecraft Education”, which in 2021 had 35 million licensed users [78]. Minecraft Education is a game-based learning platform offering teaching tools for a wide variety of subjects such as coding, engineering, design, history, and chemistry, and it enables educators to create their own tools [4, 9, 40, 47, 65, 139]. Licenses can be purchased individually or from the educational organizations for roughly US\$5–12/year. There are no advertisements or in-app purchases. While the company calls their product “license” it is really a subscription, since it is only valid for one year.

Another example is “Code Avengers”, which is an AI-supported online learning platform for teaching programming using different media including games [34]. It was developed by a Māori-owned organization with the goal to bring digital equity and job-ready skills to indigenous and underserved communities. The company sells subscriptions in the USA for US\$20-29 depending on the length of the subscription (1-12 months) [34].

Freemium: Freemium is a business model combining free services and products with premium ones. Customers are attracted by the free product/service and when they find it useful are more likely to spend money on added services. Two of the most successful examples of educational games in this segment are “Prodigy” and “Duolingo”.

“Prodigy” was originally developed for mathematics teaching (year 1-8 in school) and subsequently included English. In 2021 the game had 150 million registered users and 1 million teachers used it in the USA alone [95]. The game uses concepts from Pokémon, i.e., players can battle each other (by answering maths questions) and power up their characters, collect new characters, and explore new worlds (see Figure 1). Prodigy has a free basic version for students and schools, which results in a large user base (especially, by reducing workload to teachers). This is then monetised by offering parents a membership subscription (US\$6.25/month) with more in-game rewards for players (students) and tools for parents (regular reports, tutorials). The company also offers personalized tutorials (Zoom and for some locations in-person) for US\$25-140 per hour.

“Duolingo” is the world’s most successful gamified language learning app with more than 16 million downloads in January 2024 alone — almost ten times as much as the closest competitor [111, 116]. Duolingo uses a freemium business model: a free version with all of its learning content, and a subscription version (US\$ 6.99/month) with paywalled features such as unlimited attempts for learning activities, no advertisements, and progress tracking. The software extensively uses gamification concepts, e.g., learners can see rankings comparing their progress with other learners of the

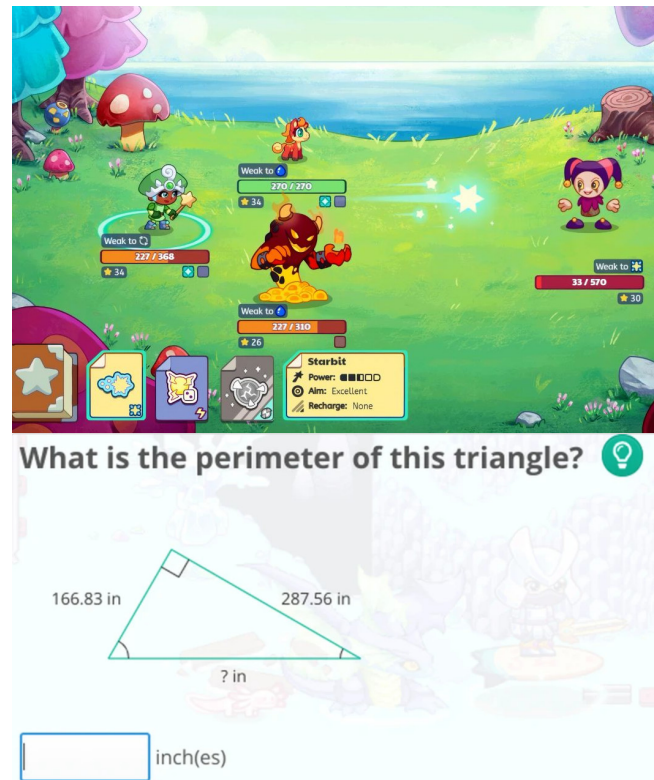


Figure 1: A screenshot of a battle in Prodigy where users have to answer mathematics questions correctly to cast spells to beat an opponent (top) and an example of a mathematics question (bottom).

same subject. In 2023 roughly 8% of users paid for a subscription, which resulted in 80% of the company’s revenue, whereas roughly 92% use the free version with ads, which contributed only 8% to the company’s income [12].

3 Games for Computing Education

Numerous games for computing education have been proposed and investigated including games from academic research and commercial products [51, 79, 126]. We provide an overview of some of the most popular approaches and then outline results from review articles about this field.

One of the earliest and arguably most successful examples of using games for computing education is “CSUnplugged”, which targets children aged 5 to 14 [15, 29]. The program started as an outreach tool but since then has been increasingly used in classrooms. It is a collection of free material that teaches Computer Science through engaging games and puzzles that involve different materials and physical activity (see Figure 2). Bell and Vahrenhold reviewed a large number of studies about the program and report mixed results ranging from improving mathematical thinking, to becoming less attracted to CS [14]. Bell suggests that the unplugged approach might work best in combination with a computer-based approach [13].

Ran and Jianyong observed several drawbacks of the unplugged approach, which seem common to interventions using non-digital media, e.g., required preparation time of teaching staff, competency differences of learners (digital tools often have automatic difficulty adjustment), too much cooperation (students can't agree on a solution approach, or some students don't actively participate), and some activities require high abstraction abilities of learners [8].

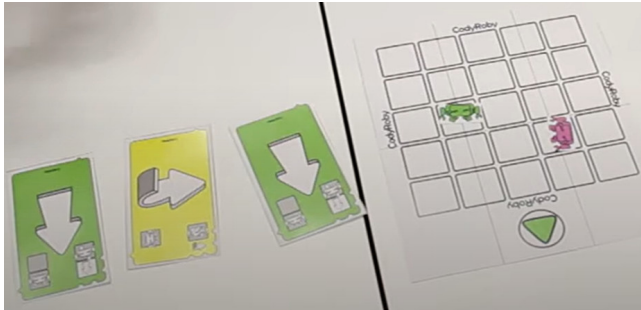


Figure 2: Example of a “CSUnplugged” activity: students “program” robots using a sequence of cards representing basic movements. Many activities are structured as simple games, e.g., players can race their robots to a target or avoid “traps”.

Hakulinen [52] used card games in a Data Structures and Algorithms course and reports a mixed but mainly positive effect on novice, but not more advanced, students' learning. The authors note that more research is needed with respect to how serious games are integrated into teaching. They also question whether computer games or non-computer games are more effective.

In 2013 the Hour of Code movement started and has since then introduced tens of millions of students in 180+ countries to computer science through one-hour coding activities often involving highly playable games like “Lightbot” [61, 134].

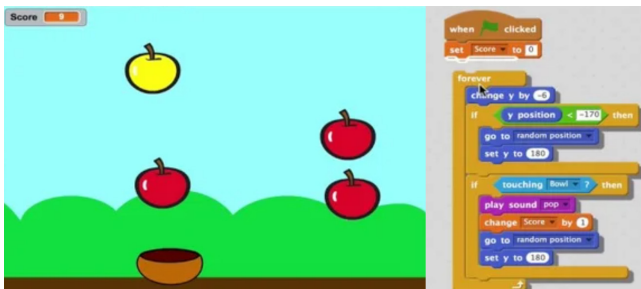


Figure 3: A simple game programmed in Scratch requiring the player to move a bowl to catch fruits.

Scratch is a popular free block-based programming language with more than 100 million registered users, that enables young learners (years 2-8 in school) to create digital stories, games, and animations using simple visual interfaces [109] (see Figure 3). It has been extended in numerous ways such as for teaching machine learning [96], integrating real-time sensor data [119], and providing formative feedback [85]. Block-based languages like Scratch do

not require learning a programming vocabulary, reduce cognitive load by chunking code into a smaller number of meaningful elements, and help users to assemble code by providing constrained direct manipulation of structure [11]. Several authors suggest that game programming using scratch empowers and motivates students and makes them more likely to continue their study [6, 87]. Disadvantages include a over-reliance on visual interfaces, code quality issues, and lack of advanced features such as recursion, memory management and advanced data structures [81].

We struggled to find games for computing education with a high replay value, i.e., where after an extended period of time the game play is still engaging. Many popular coding games, such as CodeMonkey [25] or Blockly Games [17], seem to consist of a series of small challenges that users have to solve to proceed to the next level. While these tasks can be very engaging and cover a wide variety of applications (puzzles, movies, music, etc.), we believe that once solved a level loses its attraction. CodinGame offers a wider variety of tasks including escape rooms and competitive events [26]. It is targeted at developers and also serves as a recruiting platform. Game-based learning platforms like Kahoot! offer interactive quizzes and educational games that can be used to reinforce computing concepts [129].



Figure 4: Minecraft Education enables students to develop simple games using a block-based language (right) for novices or Python or Javascript for more advanced users.

The educational game with (in our opinion) highest replay value is Minecraft Education. It offers a variety of coding lessons for school students of all age groups, where students have to solve challenges writing block-based or Python code (see Figure 4). Educators can design their own lessons, and students can create their own content in an open world environment. However, such loosely specified content can be hard to automark [55]. The tool has been highlighted as an example how social interaction and gamification should be used in coding environments for learners [47]. Kutay and Oner [65] showed that using the software resulted in a significant increase in students' knowledge of computational concepts.

Minecraft Education has several drawbacks. Having basically unlimited freedom to create virtual environments and game play, makes it easy to create material for curricula in general (history,

mathematics) [39], but makes it hard to teach specific programming constructs when using it for computing education. Minecraft Education does offer a computing curriculum [80], but it consists of specific lessons which lose their novelty after being completed. Other concerns relate to age-appropriateness, safety, required computation power, technology use, and learning generalisation [2].

3.1 Reviews of Serious Games for Computing Education

Miljanovic and Bradbury reviewed in 2018 games for learning introductory programming [79]. The authors identified 49 games (13 commercial games, 36 research games) and reported that it was difficult to independently verify learning content and that there was a lack of games that focused on data structures, development methods, and software design. The authors suggest that more research is needed on the learning benefits of different motivational strategies (competition vs. collaboration), more emphasis needs to be placed on accessibility and inclusivity, and there should be a standard for the evaluation of serious games, ideally from an independent third party.

In a more recent paper, Videnovik et al. conducted a review of game-based learning in computing education and found 114 articles published between 2017 and 2021 [126]. The authors report that the majority of papers focus on engagement (learning by playing) and there is little research on pedagogical strategy and the effectiveness of learning. The authors found that the approach is mainly used for developing computational thinking and programming skills, with more than one-third of papers about teaching object-oriented programming. While the reviewed papers indicate a wide variety of games being used, we note that most of them are based on existing platforms such as Scratch, Blockly, and Kahoot! and seem rather simplistic. This means the motivational factor seems to be the creative process of game design rather than the gameplay itself.

Villareale et al. analysed the use of reflective practice in 12 popular programming games [127]. The authors report that all 12 games offer progress displays, but only two games (“CodinGame” and “SpaceChem”) offer all four types of reflective practice, i.e., additionally process prompts (e.g., reflection on incorrect code), process models (e.g., comparison with expert solutions), and social discourse (e.g., a community library to share code).

Quiroz-Palma et al. focused on games for teaching web-programming and identified 10 such games [97]. The authors report that five of the games focused on improving programming skills. Other purposes included pre-training, improving design techniques and knowledge of programming languages, and providing exercises and assessment. Gamification techniques for engagement included challenges, rewards, virtual prizes, and points.

4 Discussion

Game-based computing education offers a variety of advantages. The most commonly reported one is enhanced engagement by making the learning process more interactive and enjoyable [130]. Other advantages are active learning and practical skill development by enabling students to practice their skills in a meaningful context, often simulating real-world scenarios [10]. By presenting

challenges and problems to solve, serious games stimulate creativity and encourage computational thinking among students [121]. Some game-based computing education tools also cultivate social interactions, e.g., by exchanging code, learning from another, or competitions requiring development of improved solutions [136].

On the negative side, there are several disadvantages such as lack of standardization [126], technological barriers, limited content [125], limited accessibility (cost, unsuitable interfaces for students with impairments, language barriers) [43], lack of pedagogical strategies [125], and being most suitable for young learners [54], which might all lead to inconsistent educational experiences. While game-based teaching tools can evaluate the correctness of students’ submissions [48], to the best of our knowledge it is not possible to customize them to specific curricula. There is also an overemphasis on entertainment making learning inefficient and potentially leading to a dilution of educational content.

4.1 Opportunities

In addition to the advantages mentioned above, we believe serious games for computing education offer a number of opportunities due to technological innovation, which so far are rarely mentioned in the context of serious games for programming.

4.1.1 Generative AI. The arguably largest opportunity is the integration of AI. Many recent papers have explored the use of LLMs (Large Language Models) for computing education including generating solutions, plain English explanations, tutorials and exercises, providing code assistance and debugging support, and performing automated grading [31, 33, 37, 42, 44, 45, 67, 69, 73, 94, 99, 104–106]. These capabilities can enable developers of serious games for programming education to generate a wider range of learning content and provide students with helpful explanations, formative feedback, and rapid assessment, which is particularly challenging and time consuming for creative tasks [56].

One of the largest problems in game development is the cost of generating 3D content. Recent advances in AI, i.e., the development of LMMs (Large Multimodal Models) and VLMs (Visual Language Models), such as GPT-4o and GPT-4o1, have made it possible to generate and assess multi-modal content [5, 42, 135]. Diffusion-based GenAI algorithms, popularised by image generators such as Dall-e, Stable Diffusion, and Mid-Journey, are now able to create convincing looking 2D and 3D assets and video content [7, 46, 68] and recent research has explored the use of LLMs to generate novel game behaviour based on player input [60]. While there are examples of entirely AI-generated 2D games and AI tools for creating such games [137], significant challenges exist, especially in the 3D domain. Examples are unrealistic animations of characters and the complexity of combining different representations of 3D content such as 3D meshes and Neural Radiance Fields (NeRFs) [98].

4.1.2 Personalisation. The term personalisation is not well defined and often interchangeably used for customisation or dynamic difficulty adjustment. We refer to it as adjusting the game to the inherent characteristics of the user, such as personality traits or cognitive abilities. While automatic difficulty adjustment based on students performance is common in many games and intelligent tutoring systems [27, 108, 140], personalisation based on our definition is

still an area of active research, often due to the lack of sensors/tools to determine inherent user traits.

We believe integrating personalisation into educational games will have significant benefits. For example, managing cognitive load is important to ensure that learning content is challenging but not too difficult [113]. Current research is investigating the use of more accessible sensors such as eye tracking for cognitive load detection [1, 59], which in combination with hardware advances, will enable developers to develop more effective learning tools for programming education [38].

Previous research has shown a correlation between spatial reasoning skills and success in STEM education including computing [18, 76, 90, 92]. In contrast to many other types of intelligence, spatial skills are malleable and training them can improve educational outcomes [18, 71, 91]. Games have been shown to train spatial skills [112, 115, 132], which suggests that there is an opportunity to develop games for computing education that improve programming, problem-solving, and spatial skills. This might also increase equity in computing education since students with low socio-economic status have statistically lower spatial skills [82].

An important motivator for using serious games is engagement. However, the effectiveness of motivational strategies depends on personality types / gamer types [16, 66, 131]. This suggests that there is an opportunity to make educational games, including games for computing education, more effective by developing gameplay and motivational strategies that adapt to the user's personality traits. However, further investigation is needed, as prior implementations of personalised education using gamer types in other educational fields have yet to demonstrate significant benefits [3, 84].

4.1.3 Virtual and Augmented Reality. Virtual Reality (VR) and Augmented Reality (AR) can be leveraged to provide an engaging experience for learners. An early review of VR for education found it had been used in a broad range of subjects, including Computer Science [62]. However, VR environments can be isolating for wearers and are unsuitable for more general classroom use where social interaction is expected. The proliferation of consumer-grade AR devices such as smart glasses has renewed interest in AR for education, although studies focused on AR for computing education have tended to focus on basic programming concepts appropriate for younger children.

Stefanidi et al. [118] demonstrated that children enjoyed using a Serious AR game to learn programming and developed spatial awareness skills. Schetz-Sobrinho et al. [107] developed a serious game to teach programming concepts such as sequences of instructions, conditions and loops. Students used the Microsoft HoloLens AR device to command a robot to achieve various tasks. Students were more motivated and more positive towards programming after playing the game. Sharma et al. [110] designed an AR serious game to teach basic programming concepts. Students used markers to program a character that would move through a 3D environment. While the game helped the players to improve their understanding of programming, it was more effective for students with lower initial programming skills.

Although CS1 and CS2 typically involve many diagrams and visual representations of computing concepts such as data storage in memory and code execution [50], and while visualisation

tools for common data structures and algorithms have been developed [128], we are yet to see widespread use of AR applications to support more interactive learning of these concepts. We see clear future opportunities for AR serious games to provide an engaging social experience for students learning these more visual aspects of computing. Tangible AR could further facilitate understanding by enabling multi-modal learning [83, 100].

4.2 Risks

In addition to the drawbacks listed at the beginning of this discussion section, we believe there are also several long-term disadvantages that have not been reported in the literature.

4.2.1 Market Dominance. The shifting from perpetual licenses to subscription and freemium models (subsection 2.3) has made the educational gaming market more accessible to smaller developers, but at the same time also makes it easier for larger developers to achieve a market dominance. Having a large user base (often encouraged by providing the software free to schools or having a basic free version) enables publishers to invest more into game content, which makes the game even more attractive to schools, advertisers, and paying premium users. A common business model in mass-market software is to offer a product for free to build a large customer base, which then can be monetised via advertisement, selling premium services, bundling with other products, partnerships, or selling data [64]. This requires massive initial investments. “Duolingo” has almost ten times as much users as the closest competing language app [111, 116], “Prodigy” seems to dominate the mathematics educational game market (used by 1 million teachers), and Minecraft Education and Scratch seem to dominate the computing education market. Having a few dominant developers results in vendor lock-in and political interference, e.g., governing bodies supporting/restricting platforms.

4.2.2 Lack of Diversity, Cultural Bias, and Inequity. The emergence of dominant developers is likely to reduce diversity due to smaller developers and diverse voices struggling to compete with established developers. Larger developers are also more likely to focus on specific markets (e.g., US curriculum) and avoid content potentially upsetting part of the customer base (e.g., LGBTQIA+ friendly). Dominant developers may unintentionally (or intentionally) embed cultural biases in their games, which can alienate or misrepresent certain groups of learners [21, 22, 101]. For example, most games we evaluated are focused on individual achievements, whereas many indigenous cultures think in terms of benefits to groups.

Due to the shift to Freemium models, there is also a risk that users from low-income families, will have a reduced learning experience. For example, while Prodigy is free to schools, a Premium subscription results in a more rewarding game play (extra awards, characters, new levels) and enables parents to track their child's progress or pay for extra online lessons.

4.2.3 Reduced Education Funding. Game-based learning might result in reduced education funding due to the initial high up-front investments, such as the cost of equipment. Additionally, there is an ongoing cost for training, support, maintenance, and security, which might mean less money for teaching staff. Technologies

(games and AI) might be promoted as a way to achieve cost savings and reduce traditional teaching [120], which will disadvantage students who are not engaged by games, lack self-efficacy and/or metacognitive skills, or have different needs.

4.2.4 Reduced Learning Outcomes. Game-based teaching tools might result in reduced learning outcomes if the emphasis is game-play rather than the educational objectives. Freemium-based business models strengthen this effect since developers often profit from maximising engagement (time spent with the tool) rather than actual learning. If the game mechanics become more engaging than the educational material, students might spend more time playing the game and less time absorbing the intended lessons. Students might perceive the game as just a source of entertainment rather than a serious learning tool. In 2021 a complaint was filed with the FTC stating that Prodigy is designed to increase playtime and make money, rather than improve learning [41].

Students might focus on winning the game rather than understanding the underlying concepts. This can lead to surface learning, where students remember facts temporarily to advance in the game but don't deeply understand or retain the information. If the game design doesn't align well with the educational goals, the learning outcomes might be compromised. For example, a game might emphasize speed or competition over comprehension and mastery of the subject matter. Games that lack clear learning objectives and structure can result in students not knowing what they are supposed to learn. Without guidance, students might miss key educational content and fail to achieve the desired learning outcomes [125].

We noticed that many educational games offer surprise gifts. Sometimes, as in the case of "Prodigy", this is restricted to premium subscribers. Since the chance of winning is related to paying money, this concept is similar to loot boxes in traditional gaming, which are sometimes considered a type of gambling [30]. We believe that there is a risk that, over time, publishers will use more mechanics from traditional games to increase player engagement and monetisation, which might further reduce learning outcomes and might even result in gaming addiction [63].

5 Conclusion & Future Work

We have reviewed the use of serious games for computing education using a holistic lens incorporating economic and historical aspects. We have shown a significant market shift from perpetual licenses to subscription and Freemium models, which are now used by all major vendors. We have discussed how this can create new risks to game-based computing education including market dominance of a few vendors, lack of diversity, cultural bias, inequity, pressure on education funding, and reduced learning outcomes. On the other hand, the combination of innovative technologies and companies with significant funding sources, opens up the possibility of creating very effective and engaging game-based learning software which is able to address current challenges in computing education, such as lack of trained staff in schools and equitable access [20].

We encourage researchers to explore the opportunities we discussed, while being aware of the risks of an increased use of game-based computing education. In particular, we encourage researchers to investigate diverse and culturally responsive teaching methods (and ideally games [93]), and to critically evaluate any push to use

a specific platform. We believe that effective game design should start with an investigation of learning barriers and followed with a co-design process investigating needs and aspirations of learners.

In future work we will explore the needs of indigenous students in New Zealand in order to develop more effective and culturally-responsive games for programming education. This is likely to involve novel and personalised motivation and learning strategies taking into account the diverse background of students.

References

- [1] Mohammad Ahmadi, Samantha W. Michalka, Sabrina Lenzoni, Marzieh Ahmadi Najafabadi, Huidong Bai, Alexander Sumich, Burkhard Claus Wünsche, and Mark Billinghurst. 2023. Cognitive Load Measurement with Physiological Sensors in Virtual Reality during Physical Activity. In *Proceedings of the 29th ACM Symposium on Virtual Reality Software and Technology (VRST '23)*. ACM, New York, NY, USA, 1–11. doi:10.1145/3611659.3615704
- [2] Omar Alawajee and Jonathan Delafield-Butt. 2021. Minecraft in Education Benefits Learning and Social Engagement. *International Journal of Game-Based Learning* 11, 4 (Oct. 2021), 19–56. doi:10.4018/IJGBL.2021100102
- [3] Coleen M Amado and Lydia S Roleda. 2020. Game element preferences and engagement of different hexad player types in a gamified physics course. In *Proceedings of the 2020 11th International Conference on E-Education, E-Business, E-Management, and E-Learning*. ACM, New York, NY, USA, 261–267.
- [4] Renate Andersen and Mikkel Rustad. 2022. Using Minecraft as an educational tool for supporting collaboration as a 21st century skill. *Computers and Education Open* 3 (2022), 1–11. doi:10.1016/j.caeo.2022.100094
- [5] Gábor Antal, Richárd Vozár, and Rudolf Ferenc. 2024. Assessing GPT-4-Vision's Capabilities in UML-Based Code Generation.
- [6] Abbas Attarwala. 2024. Engaging Middle Schoolers in Game Programming: A Scratch-Based Workshop Experience. *Journal of Computing Sciences in Colleges* 39, 6 (May 2024), 70–78.
- [7] Song Bai and Jie Li. 2024. Progress and Prospects in 3D Generative AI: A Technical Overview including 3D human. arXiv:2401.02620 [cs.AI] <https://arxiv.org/abs/2401.02620>
- [8] Ran Bao and Jianyong Chen. 2022. Characteristics and problems of unplugged computer science curriculum for young children: comparative and practical research based on the curriculum in four countries. *International Journal for Innovation Education and Research* 10, 4 (2022), 1–22. doi:10.31686/ijer.vol10.iss4.3700
- [9] David Bar-El and Kathryn E. Ringland. 2020. Crafting Game-Based Learning: An Analysis of Lessons for Minecraft Education Edition. In *Proceedings of the 15th International Conference on the Foundations of Digital Games (FDG '20)*. ACM, New York, NY, USA, 1–4. doi:10.1145/3402942.3409788
- [10] Nathalie Barz, Manuela Benick, Laura Dörrenbächer-Ulrich, and Franziska Perels. 2024. The Effect of Digital Game-Based Learning Interventions on Cognitive, Metacognitive, and Affective-Motivational Learning Outcomes in School: A Meta-Analysis. *Review of Educational Research* 94, 2 (2024), 193–227. doi:10.3102/00346543231167795
- [11] David Bau, Jeff Gray, Caitlin Kelleher, Josh Sheldon, and Franklyn Turbak. 2017. Learnable programming: blocks and beyond. *Commun. ACM* 60, 6 (May 2017), 72–80. doi:10.1145/3015455
- [12] BBC. 2024. Good, free, fun: The simple formula that has made Duolingo a daily habit for millions. <https://www.bbc.com/worklife/article/20241004-the-simple-formula-that-made-duolingo-a-daily-habit-for-millions>
- [13] Tim Bell. 2021. CS unplugged or coding classes? *Commun. ACM* 64, 5 (apr 2021), 25–27. doi:10.1145/3457195
- [14] Tim Bell and Jan Vahrenhold. 2018. CS Unplugged—How Is It Used, and Does It Work? In *Adventures Between Lower Bounds and Higher Altitudes: Essays Dedicated to Juraj Hromkovič on the Occasion of His 60th Birthday*, Hans-Joachim Böckenhauer, Dennis Komm, and Walter Unger (Eds.). Springer International Publishing, Cham, 497–521. doi:10.1007/978-3-319-98355-4_29
- [15] Tim Bell, Ian H. Witten, and Mike Fellows. 2016. *Computer Science Unplugged ... off-line activities and games for all ages*. University of Canterbury, Christchurch, New Zealand. <https://classic.csunplugged.org/documents/books/english/unplugged-book-v1.pdf>
- [16] Dan Bennett and Elisa D. Mekler. 2024. Beyond Intrinsic Motivation: The Role of Autonomous Motivation in User Experience. *ACM Transactions on Computer-Human Interaction* 5, 2 (2024), 1–40. doi:10.1145/3689044
- [17] Blockly. 2024. Blockly Games. <https://blockly.games>
- [18] Ryan Bockmon, Stephen Cooper, Jonathan Gratch, Jian Zhang, and Mohsen Dorodchi. 2020. Can Students' Spatial Skills Predict Their Programming Abilities? In *Conference on Innovation and Technology in Computer Science Education (ITI'CE '20)* (Trondheim, Norway). ACM, New York, NY, USA, 446–451.

- [19] Patrick Brantlinger. 2016. *Bread and Circuses: Theories of Mass Culture As Social Decay*. Cornell University Press, Ithaca, NY. <https://d119vjm4apzmdm.cloudfront.net/open-access/pdfs/9781501707643.pdf>
- [20] Paul Bruno, Mariam Saffar Pérez, and Colleen M. Lewis. 2022. Four Practical Challenges for High School Computer Science. <https://edpolicyinca.org/publications/4-practical-challenges-high-school-computer-science>
- [21] Christy M. Byrd. 2016. Does Culturally Relevant Teaching Work? An Examination From Student Perspectives. *Sage Open* 6, 3 (July 2016), 1–10. doi:10.1177/2158244016660744
- [22] Manuel Caingcoy. 2023. Culturally Responsive Pedagogy: A Systematic Overview. *Diversitas Journal* 8 (10 2023), 3203–3212. doi:10.48017/dj.v8i4.2780
- [23] Mario Cerezo-Pizarro, Francisco-Ignacio Revuelta-Domínguez, Jorge Guerra-Antequera, and Jairo Melo-Sánchez. 2023. The Cultural Impact of Video Games: A Systematic Review of the Literature. *Education Sciences* 13, 11 (2023), 1–21. doi:10.3390/educsci13111116
- [24] Douglas B. Clark, Brian C. Nelson, Hsin-Yi Chang, Mario Martinez-Garza, Kent Slack, and Cynthia M. D'Angelo. 2011. Exploring Newtonian mechanics in a conceptually-integrated digital game: Comparison of learning and affective outcomes for students in Taiwan and the United States. *Computers & Education* 57, 3 (2011), 2178–2195. doi:10.1016/j.compedu.2011.05.007
- [25] CodeMonkey. 2024. Coding for kids. <https://www.codemonkey.com>
- [26] CoderPad. 2024. CodinGame. <https://www.codingame.com>
- [27] Tyne Crow, Andrew Luxton-Reilly, and Burkhard Claus Wünsche. 2018. Intelligent tutoring systems for programming education: a systematic review. In *Proceedings of the 20th Australasian Computing Education Conference (ACE '18)*. ACM, New York, NY, USA, 53–62. doi:10.1145/3160489.3160492
- [28] Tyne Crow, Andrew Luxton-Reilly, Burkhard C. Wünsche, and Paul Denny. 2019. Resources and Support for the Implementation of Digital Technologies in New Zealand Schools. In *Proc. of the 21st Australasian Computing Education Conference (ACE '19)*. ACM, New York, USA, 69–78. doi:10.1145/3286960.3286969
- [29] CSUnplugged. 2024. Computer Science without a computer. <https://www.csunplugged.org/en/>
- [30] Whitney DeCamp. 2021. Loot Boxes and Gambling: Similarities and Dissimilarities in Risk and Protective Factors. *Journal of Gambling Studies* 37, 1 (2021), 189–201. doi:10.1007/s10899-020-09957-y
- [31] Paul Denny, Viraj Kumar, and Nasser Giacaman. 2023. Conversing with Copilot: Exploring Prompt Engineering for Solving CS1 Problems Using Natural Language. In *Proceedings of the 54th ACM Technical Symposium on Computer Science Education V. 1 (SIGCSE 2023)*. ACM, New York, NY, USA, 1136–1142. doi:10.1145/3545945.3569823
- [32] Paul Denny, Fiona McDonald, Ruth Empson, Philip Kelly, and Andrew Petersen. 2018. Empirical Support for a Causal Relationship Between Gamification and Learning Outcomes. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (Montreal QC, Canada) (CHI '18). ACM, New York, NY, USA, 1–13. doi:10.1145/3173574.3173885
- [33] Paul Denny, James Prather, Brett A. Becker, James Finnie-Ansley, Arto Hellas, Juho Leinonen, Andrew Luxton-Reilly, Brent N. Reeves, Eddie Antonio Santos, and Sami Sarsa. 2024. Computing Education in the Era of Generative AI. *Communications of ACM* 67, 2 (Jan. 2024), 56–67. doi:10.1145/3624720
- [34] Digital Education Ltd. 2024. Code Avengers. <https://www.codeavengers.com>
- [35] Roberto Dillon and Ori Cohen. 2013. The Evolution of Business Models in the Video Game Industry. In *Proceedings of the International Conference on Managing the Asian Century*. Springer Singapore, Singapore, 101–108.
- [36] Damien Djaouti, Julian Alvarez, Jean-Pierre Jessel, and Olivier Rampnoux. 2011. *Origins of Serious Games*. Springer London, London, 25–43. doi:10.1007/978-1-4471-2161-9_3
- [37] Jacob Doughty, Zipiao Wan, Anishka Bompelli, Jubahed Qayum, Taozhi Wang, Juran Zhang, Yujia Zheng, Aidan Doyle, Pragnya Sridhar, Arav Agarwal, Christopher Bogart, Eric Keylor, Can Kultur, Jaromir Savelka, and Majd Sakr. 2024. A Comparative Study of AI-Generated (GPT-4) and Human-crafted MCQs in Programming Education. In *Proceedings of the 26th Australasian Computing Education Conference (ACE '24)*. ACM, New York, NY, USA, 114–123. doi:10.1145/3636243.3636256
- [38] Rodrigo Duran, Albina Zavgorodniaia, and Juha Sorva. 2022. Cognitive Load Theory in Computing Education Research: A Review. *ACM Transactions on Computing Education* 22, 4, Article 40 (Sept. 2022), 27 pages. doi:10.1145/3483843
- [39] Michael O'Leary Eadaoin J. Slattery, Deirdre Butler and Kevin Marshall. 2023. Teachers' experiences of using Minecraft Education in primary school: An Irish perspective. *Irish Educational Studies* 1, 1 (2023), 1–20. doi:10.1080/03323315.2023.2185276
- [40] Travis Faas and Chaolan Lin. 2017. Self-Directed Learning in Teacher-Lead Minecraft Classrooms. In *Proceedings of the 2017 CHI Conference Extended Abstracts on Human Factors in Computing Systems (CHI EA '17)*. ACM, New York, NY, USA, 2569–2575. doi:10.1145/3027063.3053269
- [41] Fairplay. 2024. 7 reasons to say no to Prodigy. <https://fairplayforkids.org/pg/prodigy/>
- [42] Tony Haoran Feng, Paul Denny, Burkhard Claus Wünsche, Andrew Luxton-Reilly, and Steffan Hooper. 2024. More Than Meets the AI: Evaluating the performance of GPT-4 on Computer Graphics assessment questions. In *Proceedings of the 26th Australasian Computing Education Conference (ACE '24)*. ACM, New York, NY, USA, 182–191. doi:10.1145/3636243.3636263
- [43] María Fernández-Raga, Darija Aleksić, Aysun Kapucugil Ikiz, Magdalena Markiewicz, and Herbert Streit. 2023. Development of a Comprehensive Process for Introducing Game-Based Learning in Higher Education for Lecturers. *Sustainability* 15, 4 (2023), 18 pages. doi:10.3390/su15043706
- [44] James Finnie-Ansley, Paul Denny, Brett A. Becker, Andrew Luxton-Reilly, and James Prather. 2022. The robots are coming: Exploring the implications of openai codex on introductory programming. In *Proceedings of the 24th Australasian Computing Education Conference (ACE '22)*. ACM, New York, NY, USA, 10–19. doi:10.1145/3511861.3511863
- [45] James Finnie-Ansley, Paul Denny, Andrew Luxton-Reilly, Eddie Antonio Santos, James Prather, and Brett A. Becker. 2023. My AI Wants to Know if This Will Be on the Exam: Testing OpenAI's Codex on CS2 Programming Exercises. In *Proceedings of the 25th Australasian Computing Education Conference (ACE '23)*. ACM, New York, NY, USA, 97–104. doi:10.1145/3576123.3576134
- [46] Gamedevacademy.org. 2024. Generative AI Game Dev – The Complete Guide. <https://gamedevacademy.org/generative-ai-in-game-development-complete-guide/>
- [47] Yuan Garcia. 2022. What Programming IDEs Can Learn From Minecraft. In *Proc. of the 53rd ACM Technical Symposium on Computer Science Education (SIGCSE 2022)*. ACM, New York, NY, USA, 1054. doi:10.1145/3478432.3499248
- [48] Manuel J. Gomez, José A. Ruipérez-Valiente, and Félix J. García Clemente. 2023. Towards Game-based Assessment at Scale. In *Proceedings of the Tenth ACM Conference on Learning @ Scale (L@S '23)*. ACM, New York, NY, USA, 297–301. doi:10.1145/3573051.3596175
- [49] Glenda A. Gunter, Robert F. Kenny, and Erik H. Vick. 2008. Taking educational games seriously: using the RETAIN model to design endogenous fantasy into standalone educational games. *Educational Technology Research and Development* 56, 5 (2008), 511–537. doi:10.1007/s11423-007-9073-2
- [50] Philip J. Guo. 2013. Online python tutor: embeddable web-based program visualization for cs education. In *Proceeding of the 44th ACM Technical Symposium on Computer Science Education* (Denver, Colorado, USA) (SIGCSE '13). ACM, New York, NY, USA, 579–584. doi:10.1145/2445196.2445368
- [51] Thomas Hainey and Gavin Baxter. 2024. A Serious game for programming in higher education. *Computers & Education: X Reality* 4 (2024), 100061. doi:10.1016/j.cexr.2024.100061
- [52] Lasse Hakulinen. 2011. Using serious games in computer science education. In *Proc. of the 11th Koli Calling Int. Conference on Computing Education Research*. ACM, New York, NY, USA, 83–88. doi:10.1145/2094131.2094147
- [53] Siyu Han. 2022. A Study on the Psychology of Game Design Based on the Hierarchical Theory of Needs. In *Proceedings of the 2022 3rd International Conference on Mental Health, Education and Human Development (MHEHD 2022)*. Atlantis Press, Dordrecht, Netherlands, 69–78. doi:10.2991/assehr.k.220704.244
- [54] Martina Holenko Dlab and Natasa Hoic-Bozic. 2021. Effectiveness of game development-based learning for acquiring programming skills in lower secondary education in Croatia. *Education and Information Technologies* 26, 4 (2021), 4433–4456. doi:10.1007/s10639-021-10471-w
- [55] Steffan Hooper, Burkhard C. Wünsche, Andrew Luxton-Reilly, Paul Denny, and Tony Haoran Feng. 2024. Advancing Automated Assessment Tools - Opportunities for Innovations in Upper-level Computing Courses: A Position Paper. In *Proc. of the 55th ACM Technical Symposium on Computer Science Education V.1 (SIGCSE 2024)*. ACM, New York, NY, USA, 519–525. doi:10.1145/3626252.3630772
- [56] Steffan Hooper, Burkhard Claus Wünsche, Stefan Marks, Andrew Luxton-Reilly, and Paul Denny. 2024. Projects and Portfolios – An Educator's Reflections on the Summative Assessments in a Game Programming Course. In *SIGGRAPH Asia 2024 Educator's Forum*. ACM, New York, NY, USA, Article 11, 8 pages. doi:10.1145/3680533.3697071
- [57] Sarah Impey. 2023. The Cost of Entertainment: Are Games Getting Too Expensive? <https://gameanalytics.com/blog/the-cost-of-games/>
- [58] Theresia Devi Indriasari, Andrew Luxton-Reilly, and Paul Denny. 2020. Gamification of student peer review in education: A systematic literature review. *Education and Information Technologies* 25, 6 (2020), 5205–5234. doi:10.1007/s10639-020-10228-x
- [59] Peter R. Whipp James Pengelley and Anabela Malpique. 2024. A testing load: a review of cognitive load in computer and paper-based learning and assessment. *Technology, Pedagogy and Education* 1, 1 (2024), 1–17. doi:10.1080/1475939X.2024.2367517
- [60] Nicholas Jennings, Han Wang, Isabel Li, James Smith, and Bjoern Hartmann. 2024. What's the Game, then? Opportunities and Challenges for Runtime Behavior Generation. In *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology (UIST '24)*. ACM, New York, NY, USA, Article 106, 13 pages. doi:10.1145/3654777.3676358
- [61] Scott R. Bartholomew Jessica Yauney and Peter Rich. 2021. A systematic review of "Hour of Code" research. *Computer Science Education* 33, 4 (2021), 512–544. doi:10.1080/08993408.2021.2022362

- [62] Sam Kavanagh, Andrew Luxton-Reilly, Burkhard Claus Wünsche, and Beryl Plimmer. 2017. A systematic review of virtual reality in education. *Themes in science and technology education* 10, 2 (2017), 85–119.
- [63] Dongil Kim, JeeEun Karin Nam, and Changmin Keum. 2022. Adolescent Internet gaming addiction and personality characteristics by game genre. *PLoS One* 17, 2 (feb 2022), 14 pages. doi:10.1371/journal.pone.0263645
- [64] Vlad Koval. 2024. Monetizing Your Educational App. <https://medium.com/@vlad.koval/monetizing-your-educational-app-1a8d8ecfe469>
- [65] Emine Kutay and Diler Oner. 2022. Coding with Minecraft: The Development of Middle School Students' Computational Thinking. *ACM Transactions on Computing Education* 22, 2 (feb 2022), 19 pages. doi:10.1145/3471573
- [66] Dominik Lange-Nawka, Burkhard Claus Wünsche, and Samuel E. R. Thompson. 2024. That's the way I (dis-)like it! - Effect of Gamer Types and Competitiveness in VR Exergaming. *Proceedings of the ACM on Human-Computer Interaction* 8, CHI PLAY, Article 340 (Oct. 2024), 23 pages. doi:10.1145/3677105
- [67] Juho Leinonen, Arto Hellas, Sami Sarsa, Brent Reeves, Paul Denny, James Prather, and Brett A. Becker. 2023. Using Large Language Models to Enhance Programming Error Messages. In *Proceedings of the 54th ACM Technical Symposium on Computer Science Education V. 1 (SIGCSE 2023)*. ACM, New York, NY, USA, 563–569. doi:10.1145/3545945.3569770
- [68] Chenghao Li, Chaoning Zhang, Joseph Cho, Atish Waghvase, Lik-Hang Lee, Francois Rameau, Yang Yang, Sung-Ho Bae, and Choong Seon Hong. 2024. Generative AI meets 3D: A Survey on Text-to-3D in AIGC Era. arXiv:2305.06131 [cs.CV] <https://arxiv.org/abs/2305.06131>
- [69] Mark Liffiton, Brad E Sheese, Jaromir Savelka, and Paul Denny. 2023. CodeHelp: Using Large Language Models with Guardrails for Scalable Support in Programming Classes. In *Proceedings of the 23rd Koli Calling International Conference on Computing Education Research (Koli Calling '23)*. ACM, New York, NY, USA, 1–11. doi:10.1145/3631802.3631830
- [70] Richard Lord. 2024. How Monopoly came to dominate board games. <https://www.scmp.com/lifestyle/arts-culture/article/1596064/how-monopoly-came-dominate-board-games>
- [71] Tom Lowrie, Tracy Logan, Danielle Harris, and Mary Hegarty. 2018. The impact of an intervention program on students' spatial reasoning: student engagement through mathematics-enhanced learning activities. *Cognitive Research: Principles and Implications* 3, 50 (2018), 1–10. doi:10.1186/s41235-018-0147-y
- [72] Andrew Luxton-Reilly and Paul Denny. 2009. A simple framework for interactive games in CS1. In *Proceedings of the 40th ACM Technical Symposium on Computer Science Education (SIGCSE '09)* (Chattanooga, TN, USA). ACM, New York, NY, USA, 216–220. doi:10.1145/1508865.1508947
- [73] Stephen MacNeil, Andrew Tran, Arto Hellas, Joanne Kim, Sami Sarsa, Paul Denny, Seth Bernstein, and Juho Leinonen. 2023. Experiences from Using Code Explanations Generated by Large Language Models in a Web Software Development E-Book. In *Proceedings of the 54th ACM Technical Symposium on Computer Science Education V. 1 (SIGCSE 2023)*. ACM, New York, NY, USA, 931–937. doi:10.1145/3545945.3569785
- [74] Videnovik Maja, Vold Tone, Kionig Linda, Madevska Bogdanova Ana, and Trajkovic Vladimir. 2023. Game-based learning in computer science education: a scoping literature review. *International Journal of STEM Education* 10, Article 54 (2023), 23 pages. doi:10.1186/s40594-023-00447-2
- [75] Austin Albert Mardon. 2020. *The History of Board Games*. Golden Meteorite Press, Edmonton, AB, Canada.
- [76] Lauren E. Margulieux. 2019. Spatial Encoding Strategy Theory: The Relationship between Spatial Skill and STEM Achievement. In *Proc. of the Conference on International Computing Education Research (ICER '19)* (Toronto ON, Canada). ACM, New York, NY, USA, 81–90. doi:10.1145/3291279.3339414
- [77] Abraham Harold Maslow. 1943. A theory of human motivation. *Psychological Review* 50 (1943), 370–396. Issue 4. doi:10.1037/h0054346
- [78] Microsoft. 2021. Minecraft Franchise Fact Sheet April 2021. https://news.xbox.com/en-us/wp-content/uploads/sites/2/2021/04/Minecraft-Franchise-Fact-Sheet_April-2021.pdf
- [79] Michael A. Miljanovic and Jeremy S. Bradbury. 2018. A Review of Serious Games for Programming. In *Serious Games*, Stefan Göbel, Augusto Garcia-Agundez, Thomas Tregel, Minhua Ma, Jannicke Baalsrud Hauge, Manuel Oliveira, Tim Marsh, and Polona Caserman (Eds.). Springer International, Cham, 204–216.
- [80] Minecraft Education. 2024. Computing with Minecraft - Curriculum Overview. <https://www.scribd.com/document/637758579/Coding-with-Minecraft-Curriculum-Overview-1>
- [81] MIT Media Lab. 2024. Programming Concepts and Skills Supported in Scratch. <https://web.media.mit.edu/~mres/scratch/documentation/Scratch-CS-concepts.pdf>
- [82] Wenke Möhring, Andrew D. Ribner, Robin Segerer, Melissa E. Libertus, Tobias Kahl, Larissa Maria Troesch, and Alexander Grob. 2021. Developmental trajectories of children's spatial skills: Influencing variables and associations with later mathematical thinking. *Learning and Instruction* 75 (Oct. 2021), 101515.
- [83] Kyzyl Monteiro, Ritik Vatsal, Neil Chulpongsatorn, Aman Parnami, and Ryo Suzuki. 2023. Teachable Reality: Prototyping Tangible Augmented Reality with Everyday Objects by Leveraging Interactive Machine Teaching. In *Proc. of the Conference on Human Factors in Computing Systems (CHI '23)*. ACM, New York, NY, USA, Article 459, 15 pages. doi:10.1145/3544548.3581449
- [84] Alberto Mora, Gustavo F Tondello, Lennart E Nacke, and Joan Arnedo-Moreno. 2018. Effect of personalized gameful design on student engagement. In *2018 IEEE global engineering education conference (EDUCON)*. IEEE, New York, NY, USA, 1925–1933.
- [85] Florian Obermüller, Luisa Greifenstein, and Gordon Fraser. 2023. Effects of Automated Feedback in Scratch Programming Tutorials. In *Proceedings of the 2023 Conference on Innovation and Technology in Computer Science Education (ITICSE 2023)* V. 1. ACM, New York, NY, USA, 396–402. doi:10.1145/3587102.3588803
- [86] Fidelia A. Orji, Francisco J. Gutierrez, and Julita Vassileva. 2024. Exploring the Influence of Persuasive Strategies on Student Motivation: Self-determination Theory Perspective. In *Proc. of the 19th International Conference on Persuasive Technology (PERSUASIVE 2024)*. Springer-Verlag, Berlin, Heidelberg, 222–236. doi:10.1007/978-3-031-58226-4_17
- [87] Ibrahim Ouahbia, Fatima Kaddaria, Hassane Darhmaoui, Abdelrhani Elachqara, and Soufiane Lahminea. 2015. Learning Basic Programming Concepts By Creating Games With Scratch Programming Environment. *Procedia - Social and Behavioral Sciences* 191 (2015), 1479–1482. Issue 1. doi:10.1016/j.sbspro.2015.04.224
- [88] Liuxia Pan, Ahmed Tlili, Jiaping Li, Feng Jiang, Gaojun Shi, Huiju Yu, and Junfeng Yang. 2021. How to Implement Game-Based Learning in a Smart Classroom? A Model Based on a Systematic Literature Review and Delphi Method. *Frontiers in Psychology* 12, 2 (2021), 1–13. doi:10.3389/fpsyg.2021.749837
- [89] Zhongqi Pan. 2016. Guanxi, Weiqi and Chinese Strategic Thinking. *Chinese Political Science Review* 1 (2016), 303–321. Issue 2. doi:10.1007/s41111-016-0015-1
- [90] Jack Parkinson. 2020. Investigating Spatial Skills in Computing Education. In *Proc. of the Conference on International Computing Education Research (ICER '20)* (Virtual Event, New Zealand). ACM, New York, NY, USA, 340–341. doi:10.1145/3372782.3407109
- [91] Jack Parkinson and Quintin Cutts. 2020. The Effect of a Spatial Skills Training Course in Introductory Computing. In *Proc. of the Conference on Innovation and Technology in Computer Science Education (ITICSE '20)* (Trondheim, Norway). ACM, New York, NY, USA, 439–445. doi:10.1145/3341525.3387413
- [92] Jack Parkinson, Quintin Cutts, and Steve Draper. 2020. Relating Spatial Skills and Expression Evaluation. In *United Kingdom; Ireland Computing Education Research Conference (UKICER '20)* (Glasgow, United Kingdom). ACM, New York, NY, USA, 17–23. doi:10.1145/3416465.3416473
- [93] William Christopher Payne, Yoav Bergner, Mary Etta West, Carlie Sharp, R. Benjamin Shapiro, Danielle Albers Szafr, Edd V. Taylor, and Kayla DesPortes. 2021. danceON: Culturally Responsive Creative Computing. In *Proceedings of CHI '21*. ACM, New York, NY, USA, 1–16. doi:10.1145/3411764.3445149
- [94] James Prather, Paul Denny, Juho Leinonen, Brett A. Becker, Ibrahim Albluwi, Michelle Craig, Hieke Keuning, Natalie Kiesler, Tobias Kohn, Andrew Luxton-Reilly, Stephen MacNeil, Andrew Petersen, Raymond Pettit, Brent N. Reeves, and Jaromir Savelka. 2023. The Robots Are Here: Navigating the Generative AI Revolution in Computing Education. In *Proc. of the 2023 Working Group Reports on Innovation and Technology in Computer Science Education (ITICSE-WGR '23)*. ACM, New York, NY, USA, 108–159. doi:10.1145/3623762.3633499
- [95] Prodigy. 2021. 10 Years of Helping Every Student Love Learning! <https://www.prodigygame.com/main-en/blog/prodigy-education-turns-10/>
- [96] Patricio Quiroz and Francisco J. Gutierrez. 2024. Scratch-NB: A Scratch Extension for Introducing K-12 Learners to Supervised Machine Learning. In *Proceedings of the 55th ACM Technical Symposium on Computer Science Education (SIGCSE 2024)* V.1. ACM, New York, NY, USA, 1077–1083. doi:10.1145/3626252.3630920
- [97] Patricia Quiroz-Palma, Alex Santamaria-Philco, John Cevallos-Macias, Jorge Herrera-Tapia, Jorge Pincay-Ponce, Klever Delgado-Reyes, and Dolores Muñoz-Verduga. 2023. Serious game in web programming learning: A systematic literature review. In *Proc. of the 18th Iberian Conference on Information Systems and Technologies (CISTI 2023)*. IEEE, New York, NY, USA, 1–6. doi:10.23919/CISTI58278.2023.10211695
- [98] Marie-Julie Rakotosaona, Fabian Manhardt, Diego Martin Arroyo, Michael Niemeyer, Abhijit Kundu, and Federico Tomba. 2024. NeRFMeshing: Distilling Neural Radiance Fields into Geometrically-Accurate 3D Meshes. In *International Conference on 3D Vision (3DV 2024)*. IEEE, New York, NY, USA, 1156–1165. doi:10.1109/3DV62453.2024.00093
- [99] Brent Reeves, Sami Sarsa, James Prather, Paul Denny, Brett A. Becker, Arto Hellas, Bailey Kimmel, Garrett Powell, and Juho Leinonen. 2023. Evaluating the Performance of Code Generation Models for Solving Parsons Problems With Small Prompt Variations. In *Proceedings of the 2023 Conference on Innovation and Technology in Computer Science Education V. 1 (ITICSE 2023)*. ACM, New York, NY, USA, 299–305. doi:10.1145/3587102.3588805
- [100] Dmitry Resnyansky. 2020. Augmented Reality-Supported Tangible Gamification for Debugging Learning. In *IEEE International Conference on Teaching, Assessment, and Learning for Engineering (TALE 2020)*. IEEE, New York, NY, USA, 377–383. doi:10.1109/TALE48869.2020.9368410

- [101] Gabriela T Richard. 2017. Video Games, Gender, Diversity, and Learning as Cultural Practice: Implications for Equitable Learning and Computing Participation Through Games. *Educational Technology* 57, 2 (2017), 36–43.
- [102] Richard M. Ryan and Edward L. Deci. 2017. *Self-Determination Theory: Basic Psychological Needs in Motivation, Development, and Wellness*. Guilford Publications, New York, NY, USA.
- [103] Chamindi Kavindya Samarasekara, Claudia Ott, and Anthony Robins. 2024. A framework identifying challenges & solutions for high school computing. *Education and Information Technologies* 29, 1 (2024), 15621–15654. doi:10.1007/s10639-023-12329-9
- [104] Sami Sarsa, Paul Denny, Arto Hellas, and Juho Leinonen. 2022. Automatic Generation of Programming Exercises and Code Explanations Using Large Language Models. In *Proceedings of the 2022 ACM Conference on International Computing Education Research - Volume 1 (ICER '22)*. ACM, New York, NY, USA, 27–43. doi:10.1145/3501385.3543957
- [105] Jaromir Savelka, Arav Agarwal, Marshall An, Chris Bogart, and Majd Sakr. 2023. Thrilled by Your Progress! Large Language Models (GPT-4) No Longer Struggle to Pass Assessments in Higher Education Programming Courses. In *Proceedings of the 2023 ACM Conference on International Computing Education Research - Volume 1 (ICER '23)*. ACM, New York, NY, USA, 78–92. doi:10.1145/3568813.3600142
- [106] Jaromir Savelka, Arav Agarwal, Christopher Bogart, Yifan Song, and Majd Sakr. 2023. Can Generative Pre-trained Transformers (GPT) Pass Assessments in Higher Education Programming Courses?. In *Proceedings of the 2023 Conference on Innovation and Technology in Computer Science Education V. 1 (ITIcSE 2023)*. ACM, New York, NY, USA, 117–123. doi:10.1145/3587102.3588792
- [107] Santiago Schez-Sobrinio, David Vallejo, Carlos Glez-Morcillo, Miguel Á. Redondo, and José Jesús Castro-Schez. 2020. RoboTIC: A serious game based on augmented reality for learning programming. *Multimedia Tools and Applications* 79 (2020), 34079–34099. doi:10.1007/s11042-020-09202-z
- [108] Anan Schütt, Tobias Huber, İlhan Aslan, and Elisabeth André. 2023. Fast dynamic difficulty adjustment for intelligent tutoring systems with small datasets. In *Proc. of the 16th International Conference on Educational Data Mining, 11-14 July 2023, Bengaluru, India*, Mingyu Feng, Tanja Käser, and Partha Talukdar (Eds.). International Educational Data Mining Society, Massachusetts, USA, 482–489.
- [109] Scratch Foundation. 2024. 2023 Annual Report. <https://www.scratchfoundation.org/annualreport>
- [110] Vandit Sharma, Kaushal Kumar Bhagat, Huai-Hsuan Huang, and Nian-Shing Chen. 2022. The design and evaluation of an AR-based serious game to teach programming. *Computers & Graphics* 103 (2022), 1–18. doi:10.1016/j.cag.2022.01.002
- [111] Mitchell Shortt, Shantanu Tilak, Irina Kuznetcova, Bethany Martens, and Babatunde Akinkuolie. 2021. Gamification in mobile-assisted language learning: a systematic review of Duolingo literature from public release of 2012 to early 2020. *Computer Assisted Language Learning* 36, 3 (2021), 517–554. doi:10.1080/09588221.2021.1933540
- [112] Valerie J. Shute, Matthew Ventura, and Fengfeng Ke. 2015. The power of play: The effects of Portal 2 and Lumosity on cognitive and noncognitive skills. *Computers & Education* 80, 1 (2015), 58–67. doi:10.1016/j.compedu.2014.08.013
- [113] Alexander Skulmowski and Kate Man Xu. 2022. Understanding Cognitive Load in Digital and Online Learning: a New Perspective on Extraneous Cognitive Load. *Educational Psychology Review* 34, 1 (2022), 517–554. doi:10.1007/s10648-021-09624-7
- [114] Soocial. 2024. Gamification in Education Statistics. <https://www.soocial.com/gamification-in-education-statistics/>
- [115] Ian Spence and Jing Feng. 2010. Video Games and Spatial Cognition. *Review of General Psychology* 14, 2 (2010), 92–104. doi:10.1037/a0019491
- [116] statista. 2024. Leading language learning apps worldwide in January 2024, by downloads. <https://www.statista.com/statistics/1239522/top-language-learning-apps-downloads/>
- [117] statista. 2024. Video Games - Worldwide. <https://www.statista.com/outlook/dmo/digital-media/video-games/worldwide>
- [118] Evropi Stefanidi, Maria Korozi, Asterios Leonidis, Dimitrios Arampatzis, Margherita Antona, and George Papagiannakis. 2021. When Children Program Intelligent Environments: Lessons Learned from a Serious AR Game. In *Proceedings of the 20th Annual ACM Interaction Design and Children Conference (Athens, Greece) (IDC '21)*. ACM, New York, NY, USA, 375–386. doi:10.1145/3459990.3462463
- [119] Hussel Suriyaarachchi, Paul Denny, and Suranga Nanayakkara. 2022. Scratch and Sense: Using Real-Time Sensor Data to Motivate Students Learning Scratch. In *Proceedings of the 53rd ACM Technical Symposium on Computer Science Education (SIGCSE 2022) - Volume 1*. ACM, New York, NY, USA, 983–989. doi:10.1145/3478431.3499316
- [120] The Economist. 2014. Creative destruction, A cost crisis, changing labour markets and new technology will turnan old institution on its head. <https://www.economist.com/leaders/2014/06/28/creative-destruction>
- [121] Ruochen Tian, Daisuke Saito, Hironori Washizaki, Yoshiaki Fukazawa, Hiroshi Kobayashi, and Ayumi Tsuji. 2024. Enhancing Programming Education through Game-Based Learning: Design and Implementation of a Puyo Puyo-Inspired Teaching Tool. In *Proceedings of the 55th ACM Technical Symposium on Computer Science Education V.2 (SIGCSE 2024)*. ACM, New York, NY, USA, 1838–1839. doi:10.1145/3626253.3635477
- [122] April Tyack and Elisa D. Mekler. 2024. Self-Determination Theory and HCI Games Research: Unfulfilled Promises and Unquestioned Paradigms. *ACM Transactions on Computer-Human Interaction* 31, 3, Article 40 (Aug. 2024), 74 pages. doi:10.1145/3673230
- [123] Joost van Dreunen. 2011. A Business History of Video Games: Revenue Models from 1980 to Today. In *Proceedings of Cfp: The Game behind the Video Game: Business, Regulation, and Society in the Gaming Industry*. DIGRA, Finland, 1–11.
- [124] Kirk Vanacore, Adam Sales, Allison Liu, and Erin Ottmar. 2023. Benefit of Gamification for Persistent Learners: Propensity to Replay Problems Moderates Algebra-Game Effectiveness. In *Proceedings of the Tenth ACM Conference on Learning @ Scale (L@S '23)*. ACM, New York, NY, USA, 164–173. doi:10.1145/3573051.3593395
- [125] Maja Videnovik, Ana Madevska Bogdanova, and Vladimir Trajkovik. 2024. Game-based learning approach in computer science in primary education: A systematic review. *Entertainment Computing* 48 (2024), 100616. doi:10.1016/j.entcom.2023.100616
- [126] Maja Videnovik, Tone Vold, Linda Kjøning, Ana Madevska Bogdanova, and Trajkovik. 2023. Game-based learning in computer science education: a scoping literature review. *International Journal of STEM Education* 10, 1 (2023), 1–23. doi:10.1186/s40594-023-00447-2
- [127] Jennifer Villareale, Colan F. Biemer, Magy Seif El-Nasr, and Jichen Zhu. 2020. Reflection in Game-Based Learning: A Survey of Programming Games. In *Proceedings of the 15th International Conference on the Foundations of Digital Games (FDG '20)*. ACM, New York, NY, USA, 1–9. doi:10.1145/3402942.3403011
- [128] VisuAlgo.net. 2024. visualising data structures and algorithms through animation. <https://visualgo.net/en>
- [129] Alf Inge Wang and Rabail Tahir. 2020. The effect of using Kahoot! for learning – A literature review. *Computers & Education* 149 (2020), 103818. doi:10.1016/j.compedu.2020.103818
- [130] Liang-Hui Wang, Bing Chen, Gwo-Jen Hwang, Jue-Qi Guan, and Yun-Qing Wang. 2022. Effects of digital game-based STEM education on students' learning achievement: a meta-analysis. *International Journal of STEM Education* 9, 1 (2022), 1–13. doi:10.1186/s40594-022-00344-0
- [131] Zixuan Wang and Burkhard Claus Wünsche. 2024. Space War: Investigating the Impact of a Loss Aversion Strategy and Personality Traits in a VR Rowing Exergame. In *Proc. of the 2024 Australasian Computer Science Week (ACSW '24)*. ACM, New York, NY, USA, 130–139. doi:10.1145/3641142.3641173
- [132] Helen Wauk, Brian S. Woodard, Ziang Xiao, Tiffany Wenting Li, and Brian P. Bailey. 2020. A Data-Driven, Player-Centric Approach to Evaluating Spatial Skill Training Games. In *Proceedings of the Annual Symposium on Computer-Human Interaction in Play (CHI PLAY '20)*. ACM, New York, NY, USA, 349–361. doi:10.1145/3410404.3414255
- [133] Wikipedia. 2024. Replay Value. https://en.wikipedia.org/wiki/Replay_value
- [134] Cameron Wilson. 2015. Hour of code—a record year for computer science. *ACM Inroads* 6, 1 (Feb. 2015), 22. doi:10.1145/2723168
- [135] Tong Wu, Guandao Yang, Zhibing Li, Kai Zhang, Ziwei Liu, Leonidas Guibas, Dahua Lin, and Gordon Wetzstein. 2024. Gpt-4v (ision) is a human-aligned evaluator for text-to-3d generation. In *Proc. of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. IEEE, New York, NY, USA, 22227–22238.
- [136] Xiang Xu, Rui Liu, Qinqin Chen, and Hairu Yang. 2023. A Digital Game-based Learning Approach to Improve Students' Learning Performance of Senior High School Programming Courses. In *Proceedings of the 2023 4th International Conference on Education Development and Studies (ICEDS '23)*. ACM, New York, NY, USA, 23–28. doi:10.1145/3591139.3591155
- [137] YoYo Games. 2024. GameMaker - The Ultimate 2D Game Engine. <https://gamemaker.io/en>
- [138] Qian Zhang and Teomara Rutherford. 2022. Grade 5 Students' Elective Replay After Experiencing Failures in Learning Fractions in an Educational Game: When Does Replay After Failures Benefit Learning?. In *12th International Learning Analytics and Knowledge Conference (LAK22)*. ACM, New York, NY, USA, 98–106. doi:10.1145/3506860.3506873
- [139] Kening Zhu and Man Ho John Heun. 2017. Teaching and Learning of Chinese History in Minecraft: A Pilot Case-Study in Hong Kong Secondary Schools. In *Proceedings of the 2017 Conference on Interaction Design and Children (IDC '17)*. ACM, New York, NY, USA, 405–410. doi:10.1145/3078072.3084301
- [140] Mohammad Zohaib. 2018. Dynamic Difficulty Adjustment (DDA) in Computer Games: A Review. *Advances in Human-Computer Interaction* 2018, 1 (2018), 5681652. doi:10.1155/2018/5681652